

Adi Menon

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🌐 adimenon.me

STANDARDIZED TESTS

SAT
2026: R/W 740, Math 720

AP Scholar

Website Home 1 min pitch

EDUCATION

Pacific Collegiate School, Santa Cruz, CA Class of 2027
GPA Weighted 4.34 Unweighted 3.83

12th: AP English Literature, AP Physics C (E&M), AP Microeconomics,
AP Statistics, Topics in Design & Engineering (**Grading** in progress)

11th: AP Computer Science A (A/A), AP Physics C Mechanics (A/A), AP
United States History (A/A), AP English Language (A/A), Multivariable
Calculus Honors (A/B)

10th: AP Calculus BC (A/A), AP World History (A/A), AP Spanish (A/B),
Chemistry (A/A), **English 10** (A/A), Video Production 2 (A/A)

9th: ; Art1 (A/A), Biology (A/A), English 9 (A/A), Precalculus Honors
(A/A), Spanish 3 (A/A), Video Production (A/A), World History (A/A)

University of North Dakota, Online Since Fall 2026
[Math 207: Introduction to Linear Algebra](#) (2 credits, Ongoing)

Stanford University, Stanford, CA Summer 2026
CS106EA: Exploring Artificial Intelligence (B+, # credits)

UC Scout, Online **Course** Fall 2023
JAVA140: Introduction to Java, 5 high school credits

Skills

Coding-Proficient in Java, Python

Electronics- ~~President of Electronics Club at~~
~~PCS, working on the Guitar Pedal Project.~~

Expertise- Model making, circuit design,
soldering

Robotics- Expertise- Circuitry , PCB design,
CAD, Manufacturing and testing, Tech report
writing, Software development

SUMMER ACTIVITIES

Karr Company, ~~London, UK~~ Summer 2025
~~Role-~~ Software Intern
Online software development internship working on backend for a UK
based traffic violation app

Inspirit AI, Online Course Summer 2024
Course: Artificial Intelligence Intensive for High School students

Awards and Honors

2026

International Science and Engineering Fair (ISEF) — \$500 1st Award, Patent and Trademark Office Society

\$500 1st Award, “*title*” Patent and Trademark Office Society, International Science and Engineering Fair (ISEF), 2026

CSEF - Qualified participant

Santa Cruz County Science Fair - Best In Show for Senior Division; ISEF Finalist; Outstanding Electrotechnology Experiment, IEEE; Armed Forces Award

2025-

International Science and Engineering Fair - Qualified as Observer

CSEF - Qualified participant

Santa Cruz County Science Fair
1st place, Computer Science category;
Outstanding Electrotechnology, IEEE; \$100 award, Keet Family; Exceptional Genius Award, Genius Olympiad

2024

Thermo Fisher Scientific Junior Innovator Challenge - Qualified Participant

CSEF - Qualified participant

Santa Cruz County Science Fair- 1st place, Physics category; Honorable Mention for Best in Show, US Air Force; Most Outstanding

INDEPENDENT ENGINEERING ACTIVITIES

10 hours per week, 17 weeks per year

Science Fair Projects

From 2018 to present

2026

~~Title~~—*Design and Development of an Autonomous Wildfire Suppression Firefighting Robot*

- Created a model prototype firefighting robot that moves, sense and extinguish fire in rough terrain.

- Concept creator
- YOLOv8, sensors, and motion successfully detected fire and activated suppression.
- Prototyped hardware and fire protection system
- ~~Patent application in process~~
- User research conducted with Fire Marshal San Jose
- Corner case failure mode identification methods
- Protected onboard circuits from extreme heat using multilayer heat-protection materials.

2025

Title - Using Deep Learning AI to Enable the Safety of Self-Driving Cars

- Developed a purely software based project to compare two CNN models
- Used CNNs to classify road signs for safer autonomous driving.
- The dataset included 3,530 training and 638 testing images across 14 categories.
- Compared VGG-16's architecture with ResNet-50's residual network.
- ResNet-50 achieved 77% accuracy, while VGG-16 achieved only 23%.
- VGG-16 likely overfit because its 138 million parameters exceeded dataset needs.
- Both models maintained similar testing performance, indicating good generalization.

2024

Title - Towards Sensor Fusion

- Developed a multimodal sensor fusion project to precisely locate objects in the field of view
- LiDAR measured reflected laser pulses to create precise, detailed maps of surrounding objects.

Exhibit in Science, Technology Engineering and Mathematics, Yale Science and Engineering Association; Naval Science Award, Office of Naval Research; Best in Innovation and Design, Tech Interactive; Outstanding Electrotechnology Experiment, IEEE; Science Achievement Award, California Association of Professional Scientists Award

- Ultrasonic sensors measure reflected sound waves to detect nearby objects.
- Infrared sensors detect objects using reflected infrared radiation and function well in sunlight.
- Combined digital readouts from the three sensors to create a confirmation of an obstacle on the road.
- Created a test bed to calibrate XY location of the obstacle and triangulate using sensor data

2023

CSEF, 2023 - 3rd place - Physics category

Thermo Fisher Scientific Junior Innovator Challenge 2023 - Qualified Participant

Santa Cruz County Science Fair - 1st place, Physics Category; Outstanding Electrotechnology Experiment, IEEE; Science Achievement Award, California Association of Professional Scientists Award

2023

Title - Ultrasonic Eyes

- Self-driving cars developed to test HC-SR04 ultrasonic sensor's accuracy.
- Temperature, materials, and environmental disruptions were evaluated for their effects on readings.

2022

CSEF, 2025 - Qualified participant

Santa Cruz County Science Fair - 1st place, Physics category; Outstanding Electrotechnology Experiment, IEEE; Best Use of the International System of Units, US Metric Association Award

2022

Title - Flying High with Drones

2026

2026 MATE ROV World Championship - 1st place for Hephaestus Robotics Team -(Category: Ranger, (Won among 47 teams from 15 countries), St Johns, Newfoundland & Labrador, Canada

2026 MATE Northern California Regional ROV Competition - 1st place for Hephaestus Robotics Team -Category: Ranger, Watsonville, CA

TEAM ENGINEERING ACTIVITIES

5 hours per week for 6 months and , ~10 hours for 2 months

X Academy, Hephaestus Robotics Team Santa Cruz, CA

Role- VP Engineering, Team Category - Ranger

- Lead a group of 3 coders , support the CTO in making ROV, built an AI detection system
- In charge of management and production of the main ROV
- Manufacturing and testing, Tech report writing, Software development

2025 MATE ROV World Championship - 3rd place for Hephaestus Robotics Team
-Category: Ranger, Alpena, Michigan, USA

2025 MATE Northern California Regional ROV Competition - 1st place for Hephaestus Robotics Team -Category: Ranger, Watsonville, CA

2024 MATE Northern California Regional ROV Competition - 2nd place for Angry Otters Team -Category: Navigator, Watsonville, CA

The Conrad Challenge - Distinguished Conrad Innovator Award

2025 - X Academy, Hephaestus Robotics Team Santa Cruz, CA
Role- Electrical Engineer/Circuit Designer , Team Category - Ranger

Custom designed the PCB's for ROV's internal electronics
Technical report writing and Circuitry

2024 - X Academy, Hephaestus Robotics Team Santa Cruz, CA
Team Category - Navigator

2023- The Tech Challenge- Represented Pacific Collegiate School at The Tech Interactive

2026 The Conrad Challenge Houston, TX (Online)

Fireguard Technology: Innovation Stage, captain/founder, team of 2

Hobbies:

Guitar - Self educated guitarist, learn new techniques (finger picking, hammer-offs, etc.) from YouTube, practice tablature of rock songs, self-entertainment, since 2016

Wood working- Amateur carpenter/artist, Carver, Knife Whittling, Sanding and figurines, Beginner at using Dremel

Tennis- Recreational player since 2019

Cooking- Ameture cook and recipe creator

Reading - Love reading and avid reader

COMMUNITY ACTIVITIES

Aegis Living - Santa Cruz, CA

May 2026- Present- 30 hours

Help elderly in an assisted living facility with tech needs and recreation hour

Teen Kitchen Project -Santa Cruz, CA

April 2023 - December 2023- 60 hours

Prepared and cooked nourishing meals for Santa Cruz County families facing cancer and other life-threatening illnesses

YouthServe - Santa Cruz, CA

June 2024 - August 2024 - 60 Hours

Volunteered for various community volunteer needs like environmental clean-ups, graffiti removal, native plant restoration, beach clean-up, Food bank food sort and packaging etc.

Pacific Collegiate School - Santa Cruz, CA

June 2023 - August 2026 - 80 Hours

Volunteered for various community and school events needs like State Park clean up, Iron Man event volunteering, School orientation event, Beach clean up, Tutoring for math, Stuff the Bus, Video recording for school events etc.

